

Datavis_report2

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```
install.packages("tidyverse")
install.packages("readr")
install.packages("ggplot2")
install.packages("dplyr")
install.packages("tidyr")

library(tidyverse)
library(readr)
library(ggplot2)
library(dplyr)
library(tidyr)

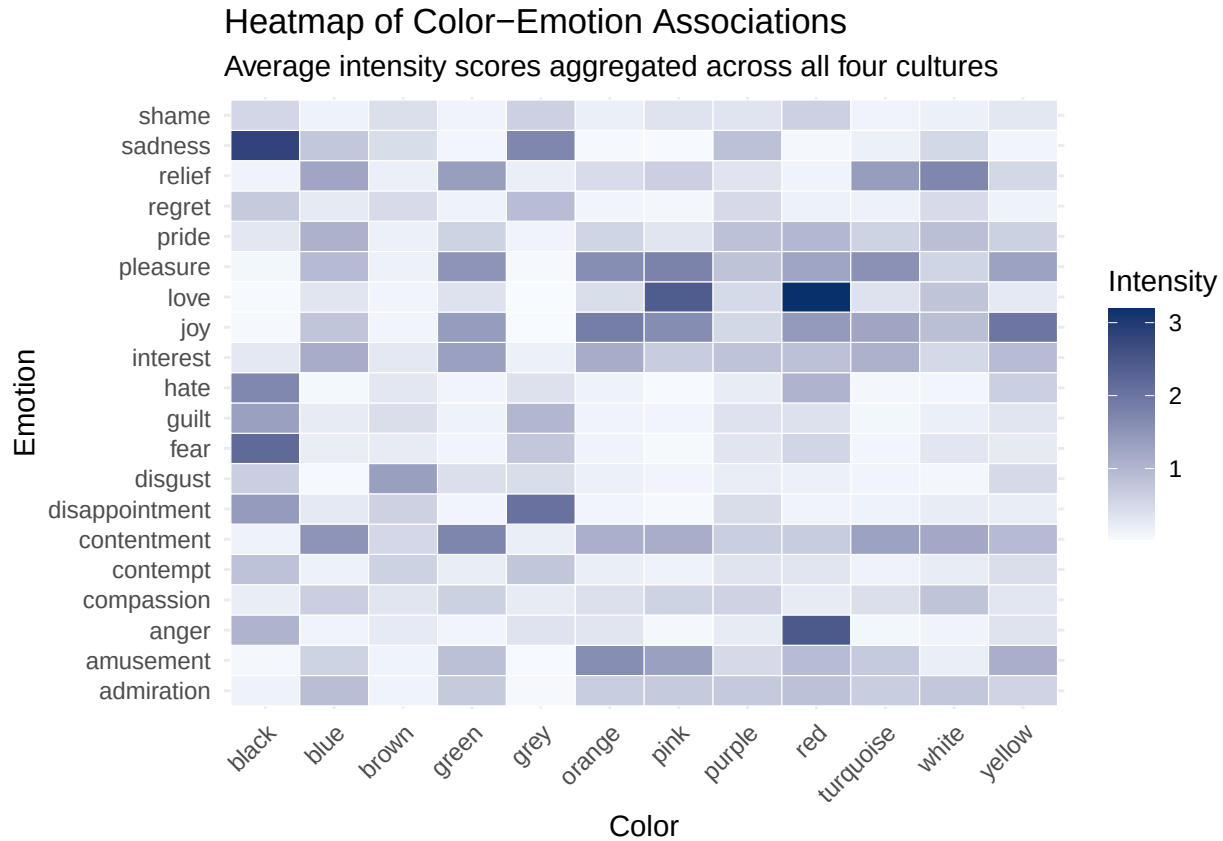
raw_data <- read_csv("color.csv")

plot_data <- raw_data %>%
  pivot_longer(
    cols = c(admiration:shame),
    names_to = "Emotion",
    values_to = "Score"
  ) %>%
  mutate(Country = case_when(
    lang == "cn" ~ "China",
    lang == "de" ~ "Germany",
    lang == "gb" ~ "UK",
    lang == "gr" ~ "Greece"
  ))

heatmap_data <- plot_data %>%
  group_by(color, Emotion) %>%
  summarise(Mean_Score = mean(Score, na.rm = TRUE))

ggplot(heatmap_data, aes(x = color, y = Emotion, fill = Mean_Score)) +
  geom_tile(color = "white") +
  scale_fill_gradient(low = "#f7fbff", high = "#08306b") +
  labs(
    title = "Heatmap of Color-Emotion Associations",
    subtitle = "Average intensity scores aggregated across all four cultures",
    x = "Color",
    y = "Emotion",
    fill = "Intensity"
  ) +
  theme_minimal() +
```

```
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
boxplot_data <- plot_data %>%
  filter(color == "white", Emotion == "sadness")

ggplot(boxplot_data, aes(x = Country, y = Score)) +

  geom_boxplot(fill = "steelblue", alpha = 0.8, outlier.shape = NA, width = 0.6) +

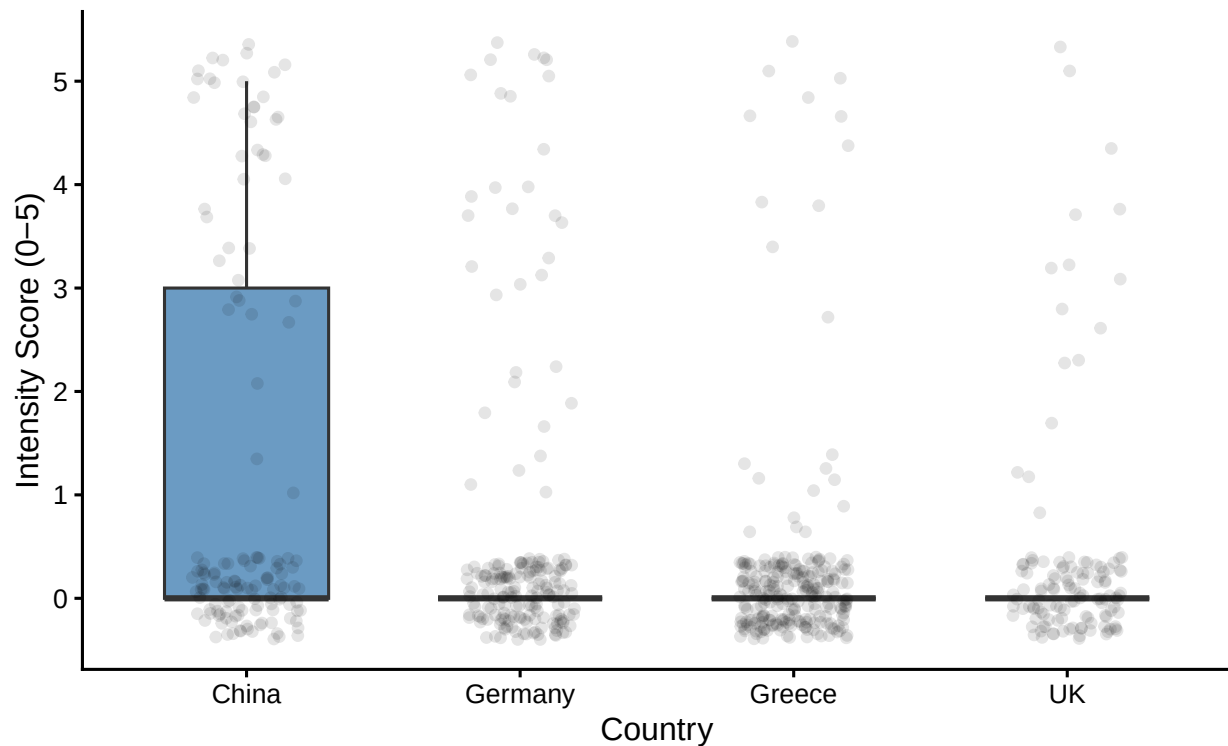
  geom_jitter(width = 0.2, alpha = 0.1, size = 1.5) +

  scale_y_continuous(breaks = c(0, 1, 2, 3, 4, 5)) +

  labs(
    title = "Cultural Divergence: White and Sadness",
    subtitle = "Comparison of Sadness intensity scores for the color White",
    x = "Country",
    y = "Intensity Score (0-5)"
  ) +
  theme_classic(base_size = 12)
```

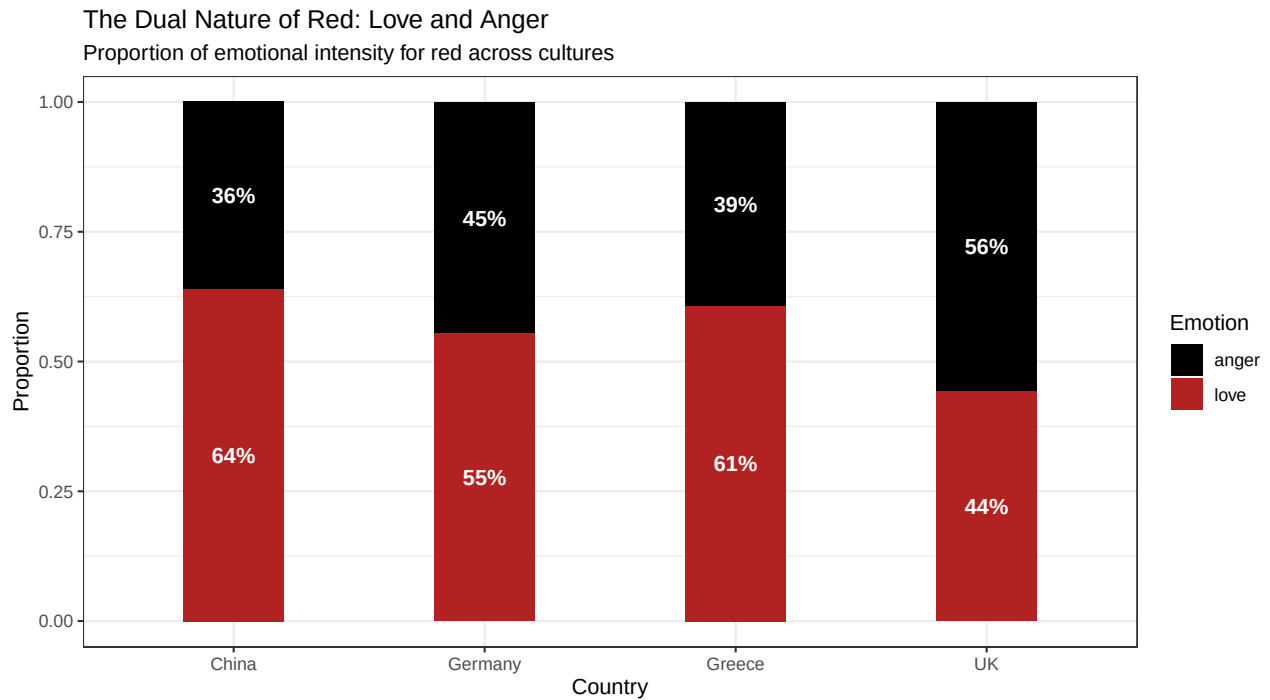
Cultural Divergence: White and Sadness

Comparison of Sadness intensity scores for the color White



```
barplot_data <- plot_data %>%
  filter(color == "red", Emotion %in% c("love", "anger")) %>%
  group_by(Country, Emotion) %>%
  summarise(Total_Score = sum(Score, na.rm = TRUE)) %>%
  mutate(Proportion = Total_Score / sum(Total_Score))

ggplot(barplot_data, aes(x = Country, y = Proportion, fill = Emotion)) +
  geom_bar(stat = "identity", position = "stack", width = 0.4) +
  scale_fill_manual(values = c("anger" = "black", "love" = "firebrick")) +
  labs(
    title = "The Dual Nature of Red: Love and Anger",
    subtitle = "Proportion of emotional intensity for red across cultures",
    x = "Country",
    y = "Proportion"
  ) +
  theme_bw() +
  geom_text(aes(label = scales::percent(Proportion, accuracy = 1)),
    position = position_stack(vjust = 0.5), color = "white", fontface = "bold")
```



To test the Universality Hypothesis, I visualized the intensity of the associations between the 12 colors and 20 emotions in the full data set. The heatmap combines the data from the four cultures and provides an overall view of the color–emotion relationships in the dataset. I computed the mean scores for all participants to create a summary table. Overall, the heatmap shows that black is mostly associated with negative emotions such as sadness, hatred, guilt, and fear, but it elicits sadness most strongly. Blue, in turn, is linked to admiration, amusement, contentment, and interest, while at the same time it can also evoke sadness. For gray and brown, the “unpleasant” emotions such as regret, disappointment, disgust, and contempt appear darker than for blue and yellow. In the red row, pleasure, love, joy, and anger are especially intense. Thus, red conveys both highly positive (pleasure/love) and highly negative (anger) emotions. Yellow is associated with emotions such as joy, amusement, and pleasure. Green appears in the cluster of relaxation, pleasure, joy, and contentment, but it also shows a slight intensity for negative emotions such as disgust, although this is much less pronounced than for red or black. In general, negative emotions cluster around black, gray, and brown, whereas positive emotions cluster around red, yellow, orange, and pink. Blue and green appear on both sides, being associated with both positive and negative emotions.

In the box plot, I compared the color white across four countries from different regions. In China, the box is very tall and wide, with a median around 2 and values ranging from 0 to 5. This indicates that Chinese participants associate white with sadness much more strongly than participants from the other countries. In Germany, Greece, and the United Kingdom, the boxes are almost stuck to zero, with medians very close to 0. The vast majority of points fall in the 0–0.5 range, with only a few slightly above.

In the bar plot, red is perceived as more positive and love-focused in China. In Germany, red is associated with love in 55% of the cases and with anger in 45%. In Greece, red is mostly a color of love and affection. In the United Kingdom, however, red is more strongly associated with anger than in the other countries.